

Supplementary Material for “Projection-Selected Multidimensional Candan Refinement after FFT Tensor Support Selection”

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The main manuscript presents the complete method and primary evidence. This supplementary material supports deeper inspection through expanded local-family ledgers, executable definitions, full-offset results, physical-parameter accuracy, quotient-stability evidence, controlled near–far metrics, and independent MATLAB replication.

I. REPRODUCIBILITY MAP

The controlled local-family campaign is reproduced by `04_experiments/eval/run_tsp_controlled_local_family_baselines.py`; its 1200 common test scenes, 480 disjoint validation scenes, row-level timings, and work contracts are stored in `05_results/tsp_controlled_local_family_paper`. Unified steady-state timing is reproduced by `run_tsp_unified_runtime_benchmark.py` and stored in `05_results/tsp_unified_runtime_paper`. Full-offset results are reproduced by `run_tsp_full_offset_sweep.py`. The controlled separation and near–far campaign is reproduced by `run_tsp_candan_stress_zeta_audit.py`. The projection protocol and runtime audits are reproduced by `run_tsp_candan_gate_protocol_audit.py` and `run_tsp_projection_gate_runtime.py`. The 9000-scene physical and ζ replay is reproduced by `run_tsp_candan_physical_zeta_audit.py`. Their registered result directories contain commands, configurations, seeds, environments, manifests, and row-level ledgers.

The independent MATLAB package in `04_experiments/matlab` implements the synthetic geometry with portable scene seeds. In particular, `exp6_candan_independent_audit.m` is the separate 1200-scene replication of the complex three-sample update. Run manifests store commands, configurations, seeds, environments, and metrics beside each output directory. All manuscript and supplementary figures are generated by `manuscript_tsp_2026/matlab_figures/generate_tsp_figures.m`; the source reads the registered CSV artifacts and exports vector PDF/SVG plus 300/600-dpi raster companions.

II. CONTROLLED LOCAL-FAMILY AND BROADER COMPARISONS

TABLE I

CONTROLLED LOCAL-FAMILY RESULTS ON 1200 COMMON TEST SCENES. INTERVALS ARE STUDENT- t 95% INTERVALS OVER FIVE SEED MEANS. IMPROVED DENOTES THE FRACTION OF SCENES WHOSE CLEAN-CHANNEL NMSE IS LOWER THAN THAT OF THE GRID ESTIMATE.

Method	NMSE [95%] (dB)	Gain (dB)	Half-bin [95%]	Improved
Grid FFT top- L	−5.43 [−5.53, −5.33]	0.00	71.9 [69.7, 74.1]%	–
Quadratic power three-sample	−6.04 [−6.13, −5.94]	0.61	71.9 [69.7, 74.2]%	100.0%
Candan complex three-sample	−29.69 [−30.33, −29.05]	24.26	79.6 [77.9, 81.3]%	100.0%
One-step fixed-support Newton	−7.71 [−7.85, −7.58]	2.28	71.7 [69.4, 74.0]%	99.4%
Cartesian local	−12.73 [−12.81, −12.65]	7.30	79.6 [77.9, 81.3]%	100.0%

All local-family methods share each noisy tensor, known L , the FFT top- L support, periodic coordinates, and joint-LS convention. Unified runtime results are reported separately in Table II to avoid mixing the common local-family ledger with the dedicated projection-selection implementation. In that audit, Candan takes 11.07 ms and projection-selected Candan takes 11.24 ms. The grid fitted energy is the sum of selected FFT-bin energies, so no second LS solve is required.

TABLE II
UNIFIED 1200-SCENE RUNTIME AUDIT (ONE THREAD, COMPLEX128, MEDIAN OF THREE REPETITIONS).

Method	Update (ms)	Decision (ms)	End to end (ms)
Candan	1.92	0.00	11.07
Explicit two-residual implementation	1.92	0.70	18.98
Projection-selected Candan	1.93	0.14	11.24

TABLE III
BROADER MATCHED ACCURACY ENDPOINTS ON THE SAME 1200-SCENE TENSOR SET.

Method	NMSE (dB)	Half-bin	Role
Grid FFT top- L + LS	-5.43	0.719	support baseline
Candan complex three-sample + LS	-29.69	0.796	matched interpolation
Cartesian local + LS	-12.73	0.796	fixed support
Complex PARAFAC-ALS	-16.85	0.524	factor reconstruction
NOMP-inspired	-53.39	0.999	iterative continuous

III. EXECUTABLE LOCAL-BASELINE DEFINITIONS

Let $X[\mathbf{k}] = \text{FFTN}(\mathcal{Y})[\mathbf{k}]$, $P[\mathbf{k}] = |X[\mathbf{k}]|^2$, and let $\mathbf{k}_\ell$ be a selected top- L bin. For axis d , the quadratic power three-sample route holds every other coordinate at $\mathbf{k}_\ell$ and uses

$$\hat{\delta}_{d,\ell} = \text{clip}_{[-1/2, 1/2]} \frac{P[\mathbf{k}_\ell - \mathbf{e}_d] - P[\mathbf{k}_\ell + \mathbf{e}_d]}{2\{P[\mathbf{k}_\ell - \mathbf{e}_d] - 2P[\mathbf{k}_\ell] + P[\mathbf{k}_\ell + \mathbf{e}_d]\}}. \quad (1)$$

Periodic indexing is used. A nonnegative or numerically flat curvature returns zero offset. After all DL offsets are formed, one joint LS fit uses the interpolated coordinates.

The proposed Candan coordinate update applies the finite-length-corrected complex three-sample ratio independently on each axis:

$$\hat{\delta}_{d,\ell}^{\text{Can}} = \text{clip}_{[-1/2, 1/2]} \frac{\tan(\pi/N_d)}{\pi/N_d} \text{Re} \left\{ \frac{X[\mathbf{k}_\ell - \mathbf{e}_d] - X[\mathbf{k}_\ell + \mathbf{e}_d]}{2X[\mathbf{k}_\ell] - X[\mathbf{k}_\ell - \mathbf{e}_d] - X[\mathbf{k}_\ell + \mathbf{e}_d]} \right\}. \quad (2)$$

It uses periodic indexing, the same selected neighborhoods, and one final joint LS fit. Equation (1) is therefore the executed quadratic power comparator, not a proxy name for Quinn or Candan.

For the one-step fixed-support Newton comparator, define $f_\ell(\mathbf{b}) = |\mathbf{a}(\mathbf{b})^H \mathbf{y}|^2$ and evaluate its analytic gradient $\mathbf{g}_\ell$ and Hessian $\mathbf{H}_\ell$ once at $\mathbf{k}_\ell$. The direction is

$$\mathbf{p}_\ell = -\mathbf{H}_\ell^\dagger \mathbf{g}_\ell, \quad \mathbf{p}_\ell \leftarrow \mathbf{p}_\ell \min \left\{ 1, \frac{0.45}{\|\mathbf{p}_\ell\|_\infty} \right\}. \quad (3)$$

The first score-improving point among $\mathbf{k}_\ell + 2^{-j} \mathbf{p}_\ell$, $j = 0, \dots, 4$, is accepted; otherwise the coarse bin is retained. Support identities remain fixed, and one joint LS fit follows the L updates.

The Cartesian route uses $\mathcal{G}_{0.45} = \{-0.45, -0.225, 0, 0.225, 0.45\}^D$ and selects

$$\hat{\mathbf{u}}_\ell = \arg \max_{\mathbf{u} \in \mathcal{G}_{0.45}} |\mathbf{a}(\mathbf{k}_\ell + \mathbf{u})^H \mathbf{y}|^2. \quad (4)$$

Separable contraction evaluates all $5^D L$ scores, followed by local joint LS. The Cartesian comparator also computes the grid joint-LS residual and returns the local result when $\rho_0 - \rho_1 \geq \tau_\rho$.

IV. FULL-OFFSET CANDAN EVALUATION

The full-offset campaign evaluates 1200 test scenes over $[-0.49, 0.49]$ bin per axis. The row-level bundle records the Candan coordinate updates, joint-LS reconstruction, per-component assignment, and five-seed accuracy intervals on the same generated scenes.

TABLE IV
FULL-OFFSET TEST ON 1200 SCENES WITH FRACTIONAL COORDINATES IN $[-0.49, 0.49]$.

Method	NMSE (dB)	Gain (dB)	Per-component hit (%)
Grid FFT top- L	-3.60	0.00	52.0
Quadratic power three-sample	-5.03	1.43	58.2
Candan complex three-sample	-18.05	14.45	65.2
One-step fixed-support Newton	-4.24	0.64	52.1
Cartesian local	-10.22	6.62	65.3

TABLE V
CANDAN COMPLEX THREE-SAMPLE COMPARATOR ON THE FROZEN 9000-SCENE CDL TEST GEOMETRY. INTERVALS USE FIVE SEED MEANS.

Profile	Scenes	Grid NMSE (dB)	Candan NMSE (dB)	Gain [95%] (dB)
CDL-A	3000	-9.65	-19.87	10.22 [10.05, 10.40]
CDL-C	3000	-9.97	-14.88	4.91 [4.58, 5.23]
CDL-D	3000	-18.29	-18.57	0.28 [0.02, 0.55]

TABLE VI
GAIN RELATIVE TO GRID FOR CALIBRATION-FREE PROJECTION SELECTION ON FIVE CDL TEST SEEDS. INTERVALS ARE PAIRED OVER SEED MEANS.

Profile	Candan (dB)	Proj.-selected (dB)	Increment [95%] (dB)	Return (%)
CDL-A	10.223	10.220	-0.003 [-0.009, 0.002]	99.8
CDL-C	4.907	4.907	-0.000 [-0.033, 0.032]	93.9
CDL-D	0.284	0.712	0.428 [0.321, 0.536]	52.2
All	5.138	5.280	0.141 [0.097, 0.186]	82.0

V. CALIBRATION-FREE PROJECTION AUDIT

The proposed rule uses the analytical decision point $\tau_\rho = 0$. Across 60 identity scenes, explicit residual and projection-energy decisions agree within 5.44×10^{-15} ; QR and reconstruction-based normalized projection energies agree within 1.09×10^{-14} . On 9000 CDL tests, the rule returns Candan in 82.0% of scenes, retains 92.9% of beneficial updates, and filters 75.7% of harmful updates.

TABLE VII
CALIBRATION-FREE PROJECTION SELECTION. INCREMENTS AND INTERVALS ARE PAIRED OVER FIVE SEED MEANS.

Profile	Candan	Selected	Increment [95%]	Return	Harmful rejection
A	10.223	10.220	-0.003 [-0.009, 0.002]	99.8	—
C	4.907	4.907	-0.000 [-0.033, 0.032]	93.9	86.5
D	0.284	0.712	0.428 [0.321, 0.536]	52.2	74.9
All	5.138	5.280	0.141 [0.097, 0.186]	82.0	75.7

VI. PHYSICAL-PARAMETER AND QUOTIENT-STABILITY EVIDENCE

The physical replay exactly reproduces all 9000 registered channel-NMSE decisions. Minimum- ζ percentiles (5th/50th/95th) are 0.015/0.111/0.445 for CDL-A, 0.017/0.165/0.637 for CDL-C, and 0.065/0.305/0.717 for CDL-D. Across the three profiles, Spearman association between ζ and clipping is -0.525 , -0.508 , and -0.506 , respectively, supporting ζ as an observable quotient-stability diagnostic.

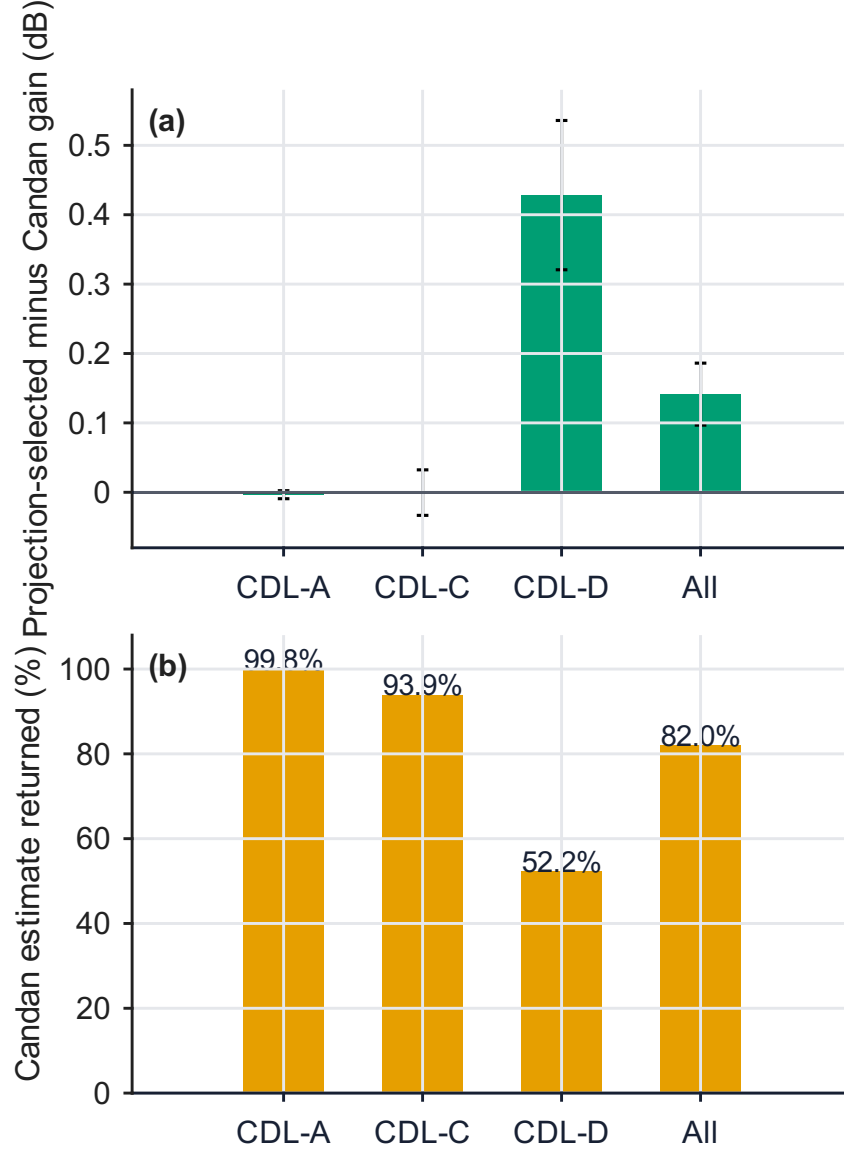


Fig. 1. Profile-level projection behavior on five CDL test seeds. (a) Paired projection-selected-minus-Candan channel-NMSE gain with seed-mean 95% intervals. (b) Fraction of scenes returning the Candan coordinates under the higher fitted-energy rule.

TABLE VIII
CDL PHYSICAL-PARAMETER RESULTS ON THE SAME 9000 SCENES.

Profile	Delay RMSE (ns)			Joint quarter-bin hit (%)		
	Grid	Candn	Selected	Grid	Candn	Selected
A	329.74	307.77	307.76	0.8	19.5	19.5
C	302.48	275.30	276.16	6.9	20.2	19.8
D	653.13	659.32	656.65	14.7	17.9	17.3
All	428.45	414.13	413.52	7.5	19.2	18.8

VII. DERIVATIVE AND CRLB SANITY CHECK

For a complex Gaussian observation with mean $\boldsymbol{\mu}(\boldsymbol{\eta})$ and covariance $\sigma^2\mathbf{I}$, the real-parameter Fisher information used by the implementation is

$$[\mathbf{F}(\boldsymbol{\eta})]_{ij} = \frac{2}{\sigma^2} \text{Re} \left\{ \left(\frac{\partial \boldsymbol{\mu}}{\partial \eta_i} \right)^H \frac{\partial \boldsymbol{\mu}}{\partial \eta_j} \right\}. \quad (5)$$

Analytic atom derivatives agree with finite differences to relative error 5.73×10^{-10} . In a separate single-component MATLAB sanity test with 200 trials per SNR, the per-axis RMSE-to- $\sqrt{\text{CRLB}}$ ratio lies in $[0.98, 1.23]$ from 10 to 30 dB, providing an independent derivative and single-component scale check for the local-refinement implementation.

VIII. CONTROLLED SEPARATION AND NEAR-FAR STRESS

The stress generator fixes $L = 4$, inserts one pair at circular Euclidean separations $\{0.25, 0.5, 1, 2\}$ bins, cycles the pair axis, and places the other components at least four bins away. Pair dynamic range is $\{0, 10, 20\}$ dB and SNR is $\{0, 10, 20, 30\}$ dB. Five seeds with five scenes per cell form the 1200-scene test, with every generated scene retained in the aggregate. Candan is positive in 48/48 cells; the mean gain is 10.46 [9.91, 11.01] dB and the minimum cell is 6.29 [4.44, 8.15] dB.

TABLE IX: Complete pure-Candan controlled near-far ledger.
Intervals use five seed means.

Sep. (bin)	DR (dB)	SNR (dB)	Gain [95%] (dB)	Per-comp. (%)	Pair (%)	Strict (%)	Strong/weak (%/%)
0.25	0	0	13.57 [8.77, 18.37]	79.0	68.0	8.0	84.0/72.0
0.25	0	10	14.30 [11.58, 17.03]	80.0	60.0	4.0	72.0/64.0
0.25	0	20	10.55 [5.66, 15.44]	75.0	52.0	4.0	68.0/64.0
0.25	0	30	13.75 [7.21, 20.29]	78.0	60.0	16.0	72.0/72.0
0.25	10	0	10.99 [9.18, 12.80]	73.0	60.0	0.0	72.0/84.0
0.25	10	10	16.42 [13.73, 19.10]	80.0	64.0	0.0	80.0/84.0
0.25	10	20	10.45 [5.53, 15.37]	70.0	60.0	0.0	72.0/84.0
0.25	10	30	14.36 [8.52, 20.20]	74.0	48.0	4.0	76.0/72.0
0.25	20	0	9.42 [7.17, 11.67]	72.0	84.0	4.0	96.0/88.0
0.25	20	10	9.06 [8.00, 10.13]	68.0	80.0	8.0	88.0/88.0
0.25	20	20	11.70 [8.34, 15.06]	77.0	96.0	0.0	100.0/96.0
0.25	20	30	9.98 [7.75, 12.21]	69.0	76.0	0.0	92.0/80.0
0.50	0	0	10.56 [6.00, 15.13]	84.0	68.0	8.0	80.0/80.0
0.50	0	10	11.86 [9.64, 14.09]	87.0	72.0	24.0	80.0/80.0
0.50	0	20	12.55 [10.36, 14.73]	88.0	80.0	16.0	88.0/80.0
0.50	0	30	10.70 [8.08, 13.32]	88.0	76.0	16.0	88.0/84.0
0.50	10	0	10.36 [8.42, 12.30]	68.0	60.0	0.0	72.0/76.0
0.50	10	10	10.68 [9.67, 11.68]	69.0	40.0	0.0	80.0/44.0
0.50	10	20	14.27 [11.78, 16.76]	77.0	44.0	0.0	80.0/60.0
0.50	10	30	12.09 [10.40, 13.77]	70.0	32.0	0.0	80.0/36.0
0.50	20	0	9.57 [8.33, 10.81]	62.0	52.0	0.0	88.0/56.0
0.50	20	10	9.19 [8.79, 9.59]	60.0	48.0	0.0	92.0/48.0
0.50	20	20	9.99 [8.15, 11.83]	59.0	44.0	0.0	80.0/56.0
0.50	20	30	9.68 [7.96, 11.40]	66.0	60.0	0.0	92.0/68.0
1.00	0	0	6.29 [4.44, 8.15]	84.0	56.0	24.0	56.0/96.0
1.00	0	10	7.17 [4.54, 9.80]	86.0	72.0	36.0	84.0/84.0
1.00	0	20	8.20 [4.97, 11.44]	89.0	80.0	28.0	84.0/88.0
1.00	0	30	6.51 [4.31, 8.71]	82.0	64.0	28.0	64.0/84.0

TABLE IX (CONTINUED)

Sep. (bin)	DR (dB)	SNR (dB)	Gain [95%] (dB)	Per-comp. (%)	Pair (%)	Strict (%)	Strong/weak (%/%)
1.00	10	0	6.77 [5.57, 7.96]	56.0	0.0	0.0	88.0/0.0
1.00	10	10	7.88 [5.08, 10.69]	60.0	0.0	0.0	92.0/0.0
1.00	10	20	7.93 [5.43, 10.43]	57.0	0.0	0.0	84.0/0.0
1.00	10	30	7.24 [5.32, 9.15]	60.0	4.0	4.0	96.0/4.0
1.00	20	0	9.36 [8.03, 10.69]	48.0	0.0	0.0	84.0/0.0
1.00	20	10	9.28 [7.17, 11.40]	48.0	0.0	0.0	92.0/0.0
1.00	20	20	9.39 [8.12, 10.65]	49.0	0.0	0.0	92.0/0.0
1.00	20	30	8.40 [7.22, 9.59]	48.0	0.0	0.0	92.0/0.0
2.00	0	0	17.51 [10.31, 24.71]	91.0	80.0	80.0	84.0/92.0
2.00	0	10	12.70 [9.13, 16.28]	85.0	68.0	68.0	76.0/88.0
2.00	0	20	16.03 [11.59, 20.46]	84.0	64.0	64.0	76.0/76.0
2.00	0	30	19.63 [13.70, 25.56]	90.0	84.0	84.0	96.0/84.0
2.00	10	0	7.19 [6.22, 8.16]	55.0	0.0	0.0	72.0/0.0
2.00	10	10	6.37 [5.17, 7.58]	54.0	0.0	0.0	84.0/0.0
2.00	10	20	7.30 [4.99, 9.62]	56.0	4.0	4.0	80.0/4.0
2.00	10	30	8.13 [4.41, 11.85]	55.0	12.0	12.0	76.0/12.0
2.00	20	0	9.28 [8.46, 10.10]	49.0	0.0	0.0	92.0/0.0
2.00	20	10	9.19 [7.90, 10.48]	48.0	0.0	0.0	92.0/0.0
2.00	20	20	9.53 [8.05, 11.00]	50.0	0.0	0.0	88.0/0.0
2.00	20	30	8.72 [8.06, 9.38]	50.0	0.0	0.0	100.0/0.0

IX. REGISTERED STABILITY AND INTEGRITY SUMMARY

TABLE X
MINIMUM- ζ DISTRIBUTIONS AND REGISTERED ASSOCIATIONS ON THE CDL REPLAY.

Profile	Scenes	$q_{0.05}$	$q_{0.50}$	$q_{0.95}$	$\rho(\zeta, \text{clipping})$
CDL-A	3000	0.015	0.111	0.445	-0.525
CDL-C	3000	0.017	0.165	0.637	-0.508
CDL-D	3000	0.065	0.305	0.717	-0.506
All	9000	0.020	0.192	0.660	-0.580

The stress and CDL replays retain every generated scene. Integrity checks confirm 48 stress cells, 1200 stress scenes, 9000 CDL scenes, and 81,000 matched physical components, with finite numerical fields throughout. The physical replay reproduces the frozen channel route with a maximum NMSE difference of 0 dB. Projection-energy and explicit-residual decisions agree to 5.44×10^{-15} , while the independently evaluated isolated Candan identity agrees to 2.8×10^{-16} across the audited apertures and fractional offsets. These checks connect the theory, estimator, tables, and MATLAB figures to registered row-level artifacts.